

Laboratory Guide 1: The Single Qubit

From Theory to Practice: First Steps with Quantum Circuits

Course: Quantum Computing Foundation

1 Introduction and Objectives

This laboratory serves as the bridge between the theoretical concepts of single-qubit states and their practical implementation using the IBM Quantum Platform.

Prerequisites:

- Lesson QC03: *The Qubit and the Bloch Sphere*
- Lesson QC04: *Single-qubit Gates*

Learning Objectives:

1. Visualize quantum states on the Bloch Sphere and Q-Sphere.
2. Implement basic gates (X, H, Z) and observe their effect on probability amplitudes.
3. Run a quantum circuit using Qiskit and interpret "Shot Noise" in measurement results.
4. Demonstrate quantum interference experimentally.

Important: Visualization Note

Bloch Sphere vs. Q-Sphere

The lectures use the **Bloch Sphere** (a single vector arrow) to represent qubits. However, the IBM Composer uses the **Q-Sphere**, which represents states as **dots** on the surface.

- **Dot Size:** Represents probability magnitude ($|\alpha|^2$).
- **Dot Color:** Represents phase (Blue = +, Red = -).

Keep this difference in mind as you proceed.

2 Part A: Visual Intuition (IBM Quantum Composer)

Platform: IBM Quantum Composer (Drag-and-Drop Interface)

2.1 Exercise 1: The Bit Flip (Pauli-X Gate)

Context: Imagine a light switch. To turn it from OFF (0) to ON (1), you apply a force. The X Gate is that force. In **QC04**, we defined: $X|0\rangle = |1\rangle$.

Student Activity

Setup: A new Composer circuit has four qubits. Reduce it to one: in the OpenQASM editor change `qubit[4] q;` to `qubit[1] q;` and `bit[4] c;` to `bit[1] c;` (or use *Edit* → *Manage registers*). With four qubits, $q_0 = 1$ is labelled 0001 and drawn one latitude below the North Pole.

Action: Drag the X gate onto the qubit line (q_0) initialized at $|0\rangle$.

Expected Effect:

- **Theory (Bloch):** The vector rotates 180 degrees to the South Pole ($|1\rangle$).
- **Probabilities:** The measurement probability for outcome "1" becomes 100%.

IBM Composer View (Q-Sphere)

What you see on screen: A single dot at the **Bottom** (South Pole).

Color: **Blue** (Phase 0).

2.2 Exercise 2: Creating Superposition (Hadamard Gate)

Context: Imagine spinning a coin. While spinning, it is a mix of Heads and Tails. The H Gate "spins" the qubit into superposition: $|+\rangle = \frac{|0\rangle + |1\rangle}{\sqrt{2}}$.

Student Activity

Action: Replace the X gate with an H gate.

Expected Effect:

- **Theory (Bloch):** The vector points to the Equator (X-axis).
- **Probabilities:** 50% for $|0\rangle$ and 50% for $|1\rangle$.

IBM Composer View (Q-Sphere)

What you see on screen: **Two Dots** (one at Top, one at Bottom).

Color: Both represent positive coefficients, so both are **Blue**.

2.3 Exercise 3: The "Invisible" Phase (Pauli-Z Gate)

Context: The Z gate creates the state $|-\rangle = \frac{|0\rangle - |1\rangle}{\sqrt{2}}$. The minus sign is a phase change, invisible to standard probability measurement.

Student Activity

Action: Place a Z gate immediately after the H gate.

Expected Effect:

- **Theory (Bloch):** The vector rotates along the equator to the negative X-axis.
- **Probabilities: Unchanged** (50/50).

IBM Composer View (Q-Sphere)

What you see on screen: Still two dots, but look at the colors!

Top Dot ($|0\rangle$): **Blue**.

Bottom Dot ($|1\rangle$): **Red**.

The Red color visualizes the negative sign (phase π) introduced by the Z gate.

3 Part B: Coding & Simulation (Qiskit)

3.1 Exercise 4: Measurement and Shot Noise

Coding Task

```
1 from qiskit import QuantumCircuit
2 from qiskit_aer import AerSimulator
3
4 qc = QuantumCircuit(1, 1)
5 qc.h(0)          # Create superposition
6 qc.measure(0, 0) # Collapse and measure
7
8 # Execute with limited shots
9 backend = AerSimulator()
10 job = backend.run(qc, shots=50)
11 counts = job.result().get_counts()
12 print(f"Results for 50 shots: {counts}")
```

Analysis: Notice that for 50 shots, you rarely get exactly 25/25. This variance is the "shot noise". Try increasing `shots` to 10000 in the notebook to see the distribution stabilize.

3.2 Exercise 5: Quantum Interference (The H-Z-H Sequence)

Coding Task

```
1 qc = QuantumCircuit(1, 1)
2 qc.h(0) # Step 1: Superposition (|+>)
3 qc.z(0) # Step 2: Phase Flip (|->)
4 qc.h(0) # Step 3: Interference
5 qc.measure(0, 0)
6
7 job = AerSimulator().run(qc, shots=1000)
8 print(job.result().get_counts())
```

Expected Result: 100% state $|1\rangle$.

Why? Even though Z was invisible in the probability chart of Exercise 3, the interference in Step 3 converts that phase difference into a measurable bit flip.

4 Part C: Assessment (Quiz)

Challenge yourself! Connect the visual results with the linear algebra from Lesson QC04.

Q1. Math Challenge: The Action of H on $|1\rangle$

In Exercise 2, we applied H to $|0\rangle$ and got $\frac{|0\rangle+|1\rangle}{\sqrt{2}}$. If we initialize the qubit to $|1\rangle$ (by adding an X gate first) and *then* apply H, what is the resulting state vector?

- A) $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$ (Two Blue dots)
- B) $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ (One Blue dot, One Red dot)
- C) $\begin{pmatrix} 0 \\ 1 \end{pmatrix}$ (One Blue dot at South)

Q2. Physics Challenge: Explaining H-Z-H

In Exercise 5, the circuit $H \rightarrow Z \rightarrow H$ acting on $|0\rangle$ produced the deterministic output $|1\rangle$. Which physical principle explains why the output is not random (50/50)?

- A) **Decoherence:** The Z gate destroyed the superposition.
- B) **Constructive/Destructive Interference:** The Z gate introduced a phase shift that caused the $|0\rangle$ terms to cancel out and the $|1\rangle$ terms to sum up during the second H gate.
- C) **Measurement Collapse:** The Z gate performed a hidden measurement.

Q3. Visualization Challenge

You are given a qubit in the state $|\Psi\rangle = \sqrt{0.3}|0\rangle - \sqrt{0.7}|1\rangle$. If you visualize this on the IBM Composer Q-Sphere, what will you see?

- A) A small Blue dot at North and a larger Red dot at South.
- B) A large Blue dot at North and a small Blue dot at South.
- C) Two equal-sized Red dots.

Solutions: Q1: B (Recall $H|1\rangle = |- \rangle$); Q2: B (Interference is key); Q3: A (Square roots determine size, minus sign determines Red color).